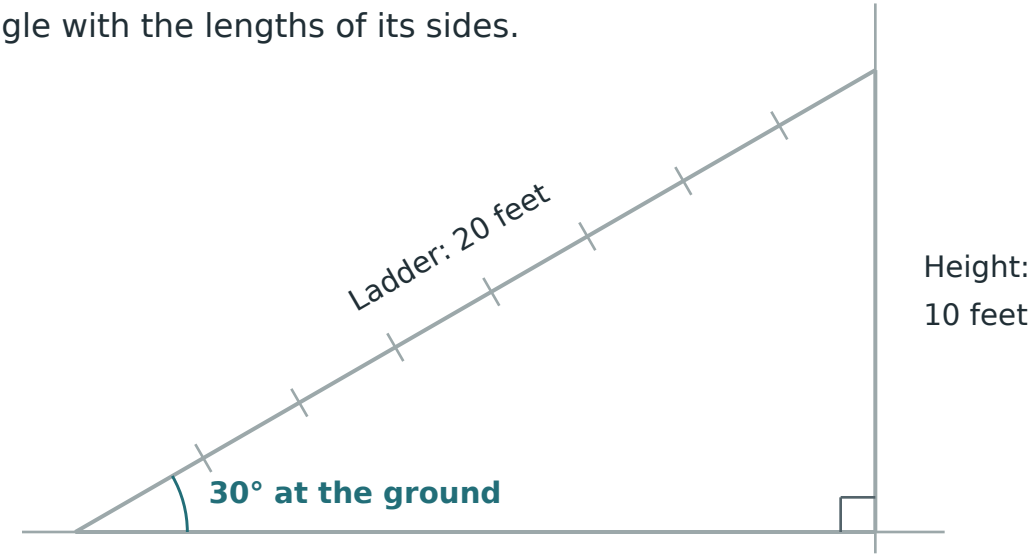
BEGIN WITH SOMETHING YOU CAN PICTURE

One ladder, three questions

Imagine a ladder leaning against a straight wall on level ground. The ladder, wall and ground make a triangle. We will keep this **same triangle** throughout the lesson.

You do not need calculus. **Trigonometry** simply helps us connect the angles of a triangle with the lengths of its sides.



Base: about 17.32 feet

First, give the parts their names.

Right triangle: a triangle with one square corner, like a sheet of paper. Here, that corner is where the wall meets the ground.

Hypotenuse (hy-POT-uh-noos): the side across from the square corner. It is always the longest side. Here, it is the ladder.

Angle: how wide two lines open apart. We measure from the ground up to the ladder: **30 degrees**. The little ° sign means “degrees.” A square corner is 90 degrees.

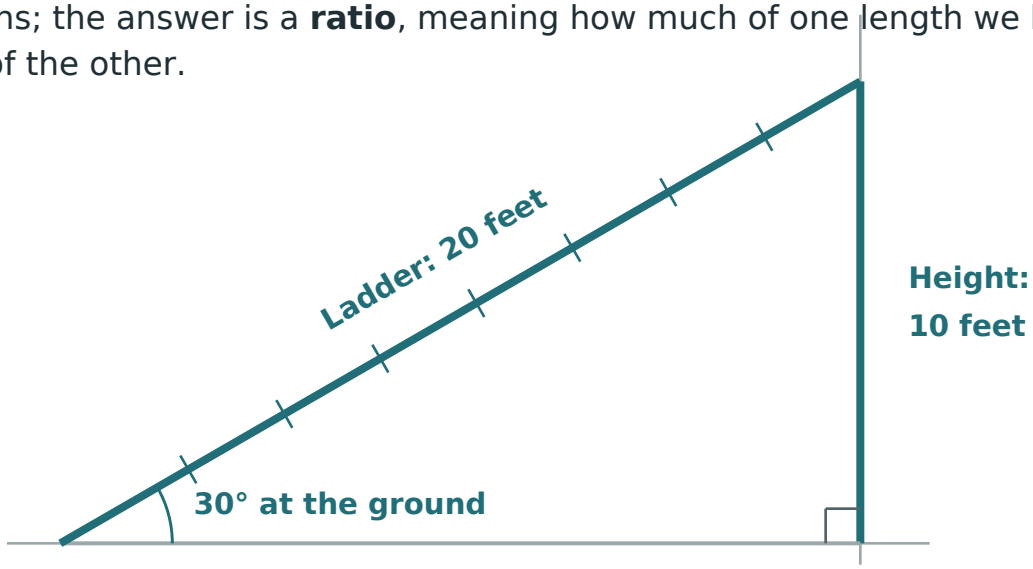
This is a paper example, not advice for safely positioning a real ladder.

QUESTION 1 / SINE — PRONOUNCED “SIGN”

How high does the ladder reach?

We know the ladder is 20 feet long and makes a 30-degree angle with the ground. We want the **height** where it touches the wall.

Sine is the height divided by the ladder length. “Divided by” compares the two lengths; the answer is a **ratio**, meaning how much of one length we have for each unit of the other.



Base: about 17.32 feet

For 30 degrees, the height is always **half the ladder length**. A calculator or a table of triangle ratios tells us this—not another measurement.

On a calculator, **sin** means sine, **=** means “equals,” and **0.5** means one half. So **sin(30°) = 0.5**. Below, ÷ means “divided by” and × means “times.”

SINE: HEIGHT DIVIDED BY LADDER LENGTH

Sine = height ÷ ladder length

Our height must be half of 20 feet:

FIND THE HEIGHT

20 feet × 0.5 = 10 feet

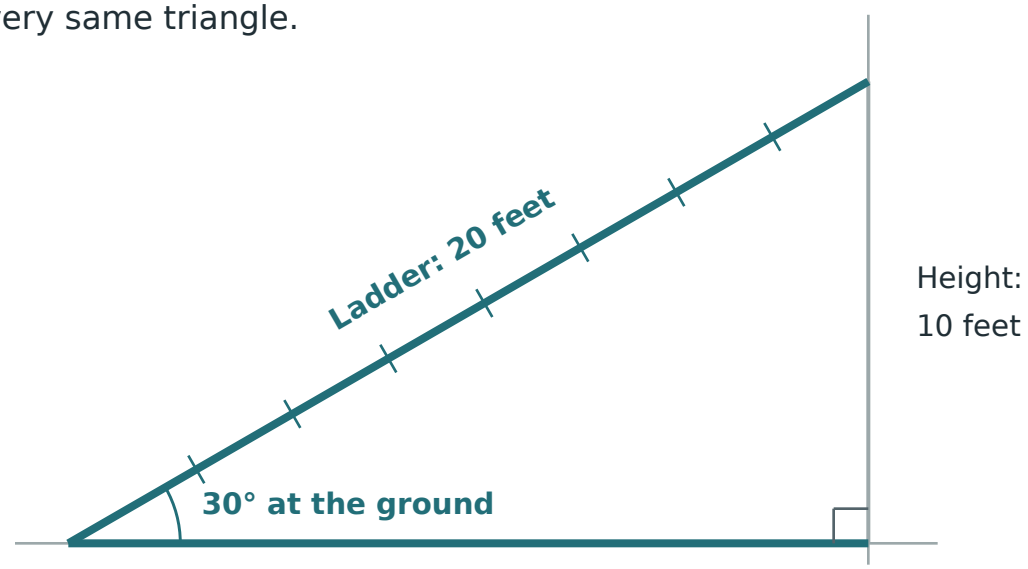
Sine is the **ratio**, not the height itself. Multiplying by the ladder length gives the height.

QUESTION 2 / COSINE — PRONOUNCED “CO-SIGN”

How far is the foot from the wall?

Keep the same ladder and the same ground angle. This time, we want the **base**: the distance along the ground from the foot of the ladder to the wall.

Cosine compares that base with the ladder length. It is a different question about the very same triangle.



For 30 degrees, a calculator gives about **0.866**: the base is about 0.866 times the ladder length. The short name is **cos**.

COSINE: BASE DIVIDED BY LADDER LENGTH

Cosine = base ÷ ladder length

To find the base, multiply the ladder length by that ratio. The symbol ≈ means “approximately equals,” or “about.”

FIND THE BASE

20 feet × 0.866 ≈ 17.32 feet

We use “about” because 0.866 is a rounded calculator answer.

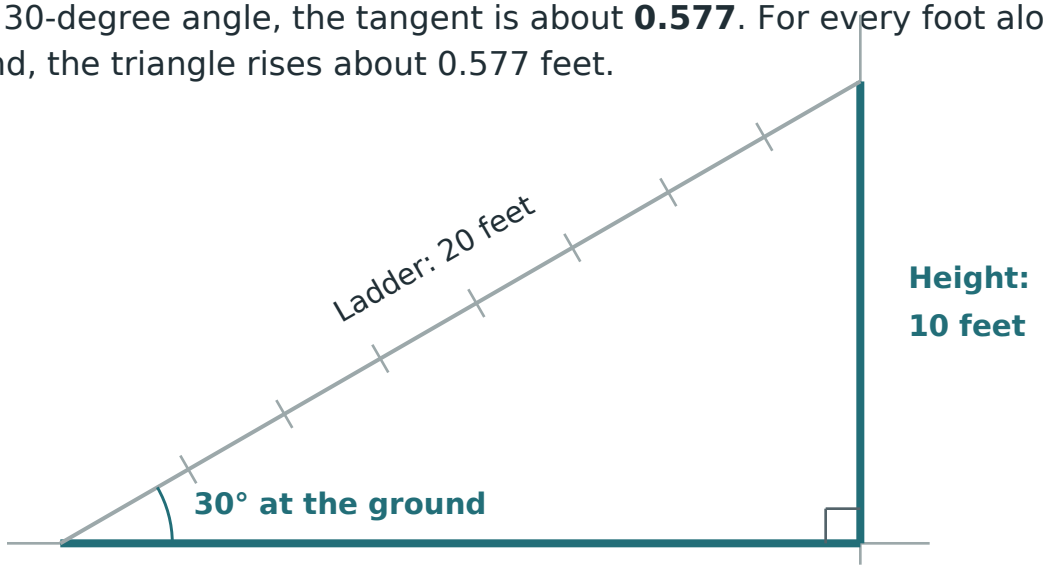
Relative to our ground angle, the height is the **opposite** side: it is across from the angle. The base is the **adjacent** side: it touches the angle and is not the hypotenuse.

QUESTION 3 / TANGENT — SHORTENED TO “TAN”

What if we know the base instead?

Imagine we know the ground angle and the distance from the wall, but not the ladder length. Can we still find the height? Yes. **Tangent** compares the height directly with the base.

For a 30-degree angle, the tangent is about **0.577**. For every foot along the ground, the triangle rises about 0.577 feet.



TANGENT: HEIGHT DIVIDED BY BASE

Tangent = height ÷ base

Our calculator writes this as **tan(30°) ≈ 0.577**. Here “tan” is just the short name of the ratio. It is not another side or another measurement.

FIND THE HEIGHT FROM THE BASE

17.32 feet × 0.577 ≈ 10 feet

Why does this work? A 40-foot ladder at 30 degrees reaches 20 feet high. Double the size; keep the shape. The height-to-ladder ratio stays one half: **10 ÷ 20 = 20 ÷ 40 = 0.5**.

All three ratios work this way: keep the angle and they stay the same, whatever the size. Change the angle and the ratios change.

Calculator tip: select **DEG** (degrees), not RAD (another angle unit, radians). Use sin, cos or tan with 30.

FROM A LADDER TO A LUNAR LANDSCAPE

A shadow makes the same triangle

Now replace the wall height with a mountain’s height. Replace the base with its shadow on level ground. A ray of sunlight takes the place of the sloping ladder.

The **Sun’s elevation angle** means how high the Sun is above the local horizon—the level direction at that place. Here we will use 30 degrees again.

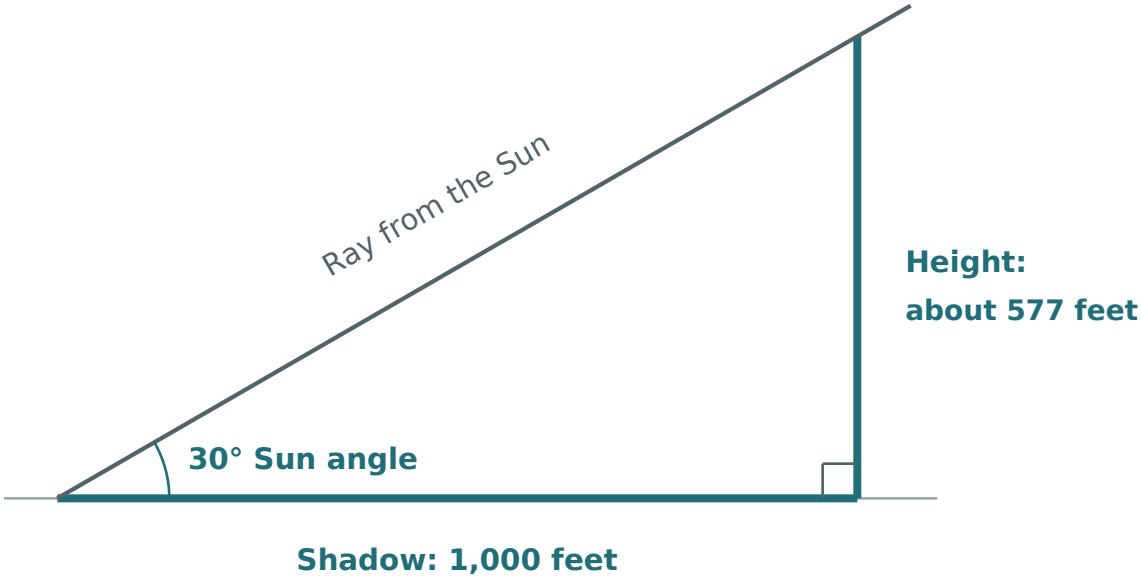


Illustration only: the upright line represents the mountain’s height.

TANGENT STILL MEANS HEIGHT DIVIDED BY BASE

Height = shadow length × tan(Sun angle)

Suppose a shadow is 1,000 feet long and the Sun is 30 degrees up. Tangent tells us the height is about 0.577 times the shadow length.

AN INVENTED EXAMPLE — NOT GALILEO’S DATA

1,000 feet × 0.577 ≈ 577 feet

This simple method also works on the Moon, approximately. We need the **true ground length** of the shadow, not an uncorrected length in a photograph, and the Sun’s angle at that lunar location.

We assume level ground and sunlight traveling in the same direction. Real measurements may need corrections for slopes, lunar curvature and the viewing angle. [2]

GALILEO'S MOON / WHAT HE ACTUALLY DID

The light reached the peaks first

In 1610, Galileo described bright mountaintops shining inside the Moon's dark region. Picture sunrise on Earth: a high peak can catch the Sun while the valley below is still dark.

He measured how far those bright peaks lay beyond the **terminator**—the boundary between lunar day and night. He compared that distance with the Moon's diameter, meaning its full width through the center.

His published calculation used the Moon's curved shape and a right triangle. He used **Pythagoras's rule**, another relationship between a right triangle's sides, to work out how far a peak had to rise to reach the sunlight.

That is not the shadow-times-tangent calculation on the previous page. Both turn light and triangles into a height, but they are different methods.

Galileo reported a height **"more than 4 Italian miles."** Those are historical Italian miles, not automatically four of today's English miles. This was his estimate—not a modern survey. [1]

TAKE THIS ONE IDEA WITH YOU

An angle fixes the triangle's proportions. One known length then tells us the scale. Each function answers a different question about the **same ladder**:

Sine: height compared with ladder length.
Cosine: base compared with ladder length.
Tangent: height compared with base.

The names are new. The idea is simply comparing lengths.

Sources & further reading

[1] Galileo, Sidereus Nuncius (1610), lunar mountains; E. S. Carlos's English translation (1880), pp. 28–30. Project Gutenberg, ebook 46036.

[2] BBC Sky at Night Magazine, "How to measure craters on the Moon" (shadow method and its limits). The source titles are clickable in this PDF.